

The Great Escape: A Classroom Activity for Teaching OLS

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Abstract

This paper describes an active learning activity for teaching ordinary least squares in an undergraduate econometrics course. In my experience, students often grasp what to do without really understanding why OLS works the way it does. In this activity, students choose an escape path across a prison courtyard, minimizing the sum of squared distances from "guard towers" in the room. This simple, fun activity gets students involved in a physical scenario with meaningful minimization consequences. I describe the activity design and what instructors need to conduct it, in the hope that others find it useful in their own classrooms.

Keywords: econometrics, ordinary least squares, active learning, regression, kinesthetic learning

JEL codes: A11, A22

Target classroom environment: undergraduate econometrics (ECON 3160). *Topic area:* OLS mechanics.

1 Introduction

Econometrics is required in most undergraduate economics programs, and gradually expanding to other programs such as business analytics and finance. Despite the virtually canonical sequence of topics taught across courses and the seemingly universal reliance on a limited pool of textbooks, outcomes among students often vary wildly across programs.

Among the factors affecting student learning, the most directly controllable for economics educators is our approach to classroom engagement. Becker and Watts (2001, 2008) find the economics profession has a longstanding reliance on "chalk and talk" lectures. Becker (2000) and several others have made the case for active learning as an alternative to these methods (Becker and Watts, 2001; Salemi, 2002; Hoyt and McGoldrick, 2012; Allgood et al., 2015; Harmon and Lambrinos, 2012; Hall, 2012; Diamond, 2009). But among the various papers addressing teaching methods, econometrics has historically received less attention than principles courses.

Angrist and Pischke (2017) argue that econometrics instruction remains abstract, preoccupied with the search for true models and with technical concerns tied to the classical regression assumptions, at the expense of the questions applied work actually focuses on. I have often felt the same about the structure of econometrics texts and courses. Wozny et al. (2018) veered away from this abstract methodology, providing active learning evidence in the field with a randomized controlled trial of a flipped classroom in introductory econometrics. This increased interest in econometrics teaching has led to the development of some interesting teaching methods (McKee and Orlov, 2021; Byker et al., 2022).

In this activity, I introduce OLS mechanics to students by having them maneuver their way through a prison courtyard. Using readily available chairs or classroom objects, I set up hypothetical guard towers. Students must navigate their escape on a straight path which serves as

our regression line, and the distance from the path to each tower is a residual. The activity is guided and I solve the setup before demonstration. This allows me to plan three paths in sequence, with the third minimizing the sum of squared residuals. Once students have identified the correct path, a final component of the activity positions three students across the room for a visual sum of squares decomposition.

Though this paper contributes to the econometrics education literature, physical demonstrations of regression have a longer history outside economics. Connor (2003) uses classroom space and students' bodies to illustrate central tendency, variability, correlation, and regression. Those activities give students a physical representation of the regression line. This activity expands on this work, giving them a physical representation of the objective function.

The activity is useful for students beyond being memorable or unique. I believe the narrative of the exercise helps students to understand regression at a fundamental level because the optimization problem is more tangible. When they think of themselves as a prisoner crossing open ground, they have a clear objective with quantifiable costs to being on the wrong path. It also provides motivation for the intuition of least squares. Students instinctively want to minimize their total exposure to the guard towers. In my experience, students have worked through the task on their own before I arrive at any formal derivation.

The next section describes the activity in detail and Section 3 provides useful implementation notes for others to use in their classrooms such as pacing, class size considerations, and spacing concerns. Section 4 concludes the paper.

2 The Activity

To prepare the activity, start by arranging the six guard towers in the room. Chairs are typically the easiest item, but feel free to get creative. When arranging, ensure enough variance between the towers so the correct path is not visually obvious. Otherwise, the exact locations are arbitrary. Since the activity is guided and the solutions determined in advance, the first two paths may be any plausible alternative to the solution.

For continuity, set up the prison perimeter to be roughly rectangular. Tower placement is done by eye and should reflect the general relationships of the distances chosen when preparing the activity. This helps to maintain the narrative and avoid confusion during the activity. You can use paced steps or invented round numbers when identifying the distances, as long as towers visibly further from the path are consistent with the assigned values.

In my demonstration, I assign towers at (1,5.5), (2,2.5), (3.5,6), (4.5,2.8), (6,5.8), and (7,3). These fall into two loose bands, so any path fitting one band poorly fits the other. Fewer towers would work. Six mirrors the perimeter of a courtyard, which is why I settled on that number, but four or five would demonstrate the same tradeoff with less arithmetic at the board. The numbers above produce a rather flat regression line, which could be useful for a myriad of economic significance discussions in its own right. However, the characteristics of the solution are entirely up to the instructor.

2.1 The Scenario

Tell students they are prisoners crossing a courtyard toward an exit on the far side. Each chair is a guard tower. The path they choose is their route across open ground, and every step of that route

is exposed to every tower at once. The closer the path runs to a tower, the greater the chance of being seen. The objective is to find the path minimizing the total squared exposure. This keeps students focused on the fact that we are adding up risk across all six towers rather than worrying about any given tower. Before starting, you can identify cardinal directions in the room or label the guard towers.

2.2 Trial and Error

Deriving the solutions beforehand is critical. If students were to search freely for the best path, the open exploration would consume the class period. Preparation ensures students converge on the correct path within the allotted time. It is also useful to have three regression lines plotted in the lesson plan. This helps when eyeing the path in the room. I use three paths in sequence, each correcting a specific flaw in the one before it.

Before attempting the escape, have the students identify the mean values of X and Y . This establishes the pivot point of the regression line at $(\bar{X}, \bar{Y})$. This is helpful visually. More importantly, it establishes the foundation for the SST decomposition later on. Since my solution is relatively flat, I make the preceding paths visually positive or negative. The comparison is useful, but it also helps to build the narrative by overcorrecting the second path. Below is a sample walkthrough of how I address the three paths for the chosen values above, with each path demonstrated by having the students lay a rope along the floor as the visual path.

Path 1 is angled steeply upward, at a slope of 0.456. The path begins in the southwest corner of the perimeter and extends to the northeast corner. It should be obvious that the path is inefficient (Figure 1). The regression line runs extremely close to two of the corner towers, but very far from the adjacent towers. A discussion on quadratic loss and overall risk follows naturally from here.

We cannot fully eliminate risk, but it is obvious that flattening the slope would result in a lower overall risk.

[Figure 1 about here]

To demonstrate, I compute Total Squared Exposure on the board, which comes to 21.53. Each subsequent path follows the same procedure, and we establish a ranking of total risk exposure throughout the activity. This is also where Total Squared Exposure is identified as Sum of Squared Residuals.

Path 2 corrects Path 1 by tilting the other way. Starting at the northwest tower, we move to the southeast tower, giving us a slope of -0.344 . The path is better overall, with a Total Squared Exposure of 15.17. Visually, it appears as though the number of guard towers our path is in close proximity to has not changed. We've essentially mirrored the problem from the first path.

[Figure 2 about here]

Before solving for the correct solution, this is a good opportunity to clarify misunderstanding with students. Some may feel like the example is counterintuitive since we want to minimize the distance between our data points and the regression line. It is important to point out that the issue is not really in our proximity to the towers, but in the inefficient allocation of total risk on the path. Being this close to the northwest tower means we are also far away from the southwest tower. If we're trying to minimize our Total Squared Exposure, there is a tradeoff between data points, and we cannot ignore the rest for the sake of one.

Path 3 brings us to our solution, at a slope of -0.094 , with Total Squared Exposure at 13.52. It balances the competing demands of six towers that cannot all be satisfied at once.

[Figure 3 about here]

The execution of the final path varies slightly from the first two. Rather than suggesting a path based on intuition or comparison, we begin the final escape attempt with a few basic calculations.

Students calculate the covariance of our variables, then our β using

$$\beta = \frac{COV(X, Y)}{VAR(X)} = \frac{-0.5}{5.3} = -0.09434$$

Next, I have them compute the mean of both variables and solve for the intercept using the formula below. Computing means, covariance, and some fundamental algebra skills are all that is needed for this exercise.

$$\hat{\alpha} = \bar{Y} - \beta\bar{X} = 4.267 + 0.09434(4) = 4.64436$$

Taking these values, we compare the Total Squared Exposure of 13.52 against the other paths (Table 1). Using the plotted lines from the lesson plan and the calculations above, students visualize the path in the room. At this point, instructors can use R, Excel, or the appropriate tool for the course to demonstrate the same computational results as well.

[Table 1 about here]

This is another opportunity for class discussion. Once the final solution has been demonstrated and students can see the comparison, it is useful to point out that the risk of being caught is still very real, only now it is minimized given the constraints of the data (guard locations). Over several escape attempts, taking this path may not always lead to a successful escape (accurate predictions), but it maximizes the likelihood of success given alternative paths (minimum squared error fit).

2.3 Total Squared Exposure

Once Path 3 is on the floor, ask students to think about detection risk at a sequence of distances from a single tower. I've often used 200 yards, 100 yards, 25 yards, and 10 yards. At 200 yards, taking one step closer doesn't meaningfully increase the risk of being spotted. At 100 yards, it may start to matter, but the distance is still great enough that movement doesn't feel constrained for the

“prisoners.” At 25 yards, and then at 10, the same single step becomes the difference between crossing the courtyard and getting caught.

At least one distance should be large enough that a step is obviously negligible and another small enough that a step is obviously decisive, with a couple of points in between so students feel the change happening gradually rather than as a jump between extremes. Any sequence with that spread works to build intuition. This helps them to understand the tradeoff between moving closer to any given data point and further from others. The marginal certainty and safety gained from moving away from a guard tower on one end is less than the marginal risk associated with an additional step towards another.

2.4 Decomposing Variation in Y

For the last part of the activity, leave the rope on the floor along Path 3 and position three students across the room. One stands on the rope, at the fitted value. One stands at a tower, at the actual value. Once they are positioned, ask the class what their best guess of the actual value would be if they had no idea where any of the towers were.

Students should arrive at the mean, though they can be directed if necessary. Position the third student at the mean of Y , and point out that the rope already crosses that point, and so did both of the paths we rejected, because the bivariate regression line passes through the sample means.

From here, identify the three values of Y and the three distinct distances created by these values. The distance from the actual value to the mean is the total deviation. The distance from the fitted value to the mean is the part the model explains. The distance from the actual value to the fitted value is what we could not explain.

[Figure 4 about here]

2.5 Activity Components

Table 2 collects the correspondences the activity relies on.

[Table 2 about here]

3 Implementation Notes

Time. The full activity fits comfortably in a 75-minute period and can be run in 50 minutes by tightening the discussion between paths. The decomposition exercise takes roughly ten minutes and should be retained if time runs short, since it is the only part that makes the sum of squares decomposition physical.

Space. Any room where six chairs or objects can be spread out and students can walk between them. The rope is optional, but most will find it useful, since a visible line on the floor is easier for the whole class to see than a path students are asked to imagine.

Class size. The activity works best in small classes if instructors want full participation. In a larger section or auditorium, running the activity at the front of the room with a small group of volunteers will suffice.

Numbers. The coordinates in Section 2 can be used directly. Instructors who prefer their own arrangement should ensure the appropriate variance between towers and recompute the three totals in advance. I recommend having the arithmetic prepared in the lesson plan, though it is simple enough instructors should have no issues otherwise.

4 Concluding Thoughts

My goal was to provide a novel and valuable method of teaching bivariate OLS mechanics. In my experience, the activity has been successful in engaging students and cracking open the "black box" of regression. But I think it extends beyond success in the classroom and demonstrates that econometrics instructors can successfully deviate from traditional lecture methods, especially with students of different mathematical backgrounds.

The activity gives students a more intuitive understanding of what regression is estimating. I think this is especially important for quantitative courses such as statistics or econometrics, where students often meet the material with a fundamental level of apprehension. Working through a physical example like this provides clarity they may not obtain otherwise, potentially building the confidence to pursue more advanced skills with greater enthusiasm. Further, engaging the students in a physical activity such as this prompts more engagement during the lesson, helping to identify the students who are unsure about the concepts who may otherwise have not asked questions. All of this has the potential to lessen the gap between student outcomes in courses like econometrics.

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Tables and Figures

Table 1: Residuals by tower under each path.

Tower	Path 1	Path 2	Path 3 (OLS)
(1, 5.5)	2.600	0.200	0.950
(2, 2.5)	-0.855	-2.455	-1.955
(3.5, 6)	1.961	1.561	1.686
(4.5, 2.8)	-1.694	-1.294	-1.419
(6, 5.8)	0.622	2.222	1.722
(7, 3)	-2.634	-0.234	-0.984
Total Squared Exposure	21.53	15.17	13.52

Note. All three paths pass through the means (4, 4.267). Slopes are 0.456, -0.344, and -0.094 respectively, with Path 3 fitted by OLS.

Table 2: Physical elements and their statistical counterparts.

Physical element	Statistical concept
Chair (guard tower)	Data point
Prisoner's path	Fitted regression line
Tower spread	Competing constraints across observations
Distance from path to tower	Residual
Path 2's overcorrection	Tradeoff imposed on the rest of the fit by improving one point
Detection risk at 200, 100, 25, 10 yards	Rationale for penalizing large deviations disproportionately
No path eliminates risk	OLS minimizes total error, it does not eliminate it
Best guess with tower locations unknown	The mean as the uninformed predictor
Rope crossing the third student	Regression line passing through the means
Distances among the three students	Decomposition of the total sum of squares

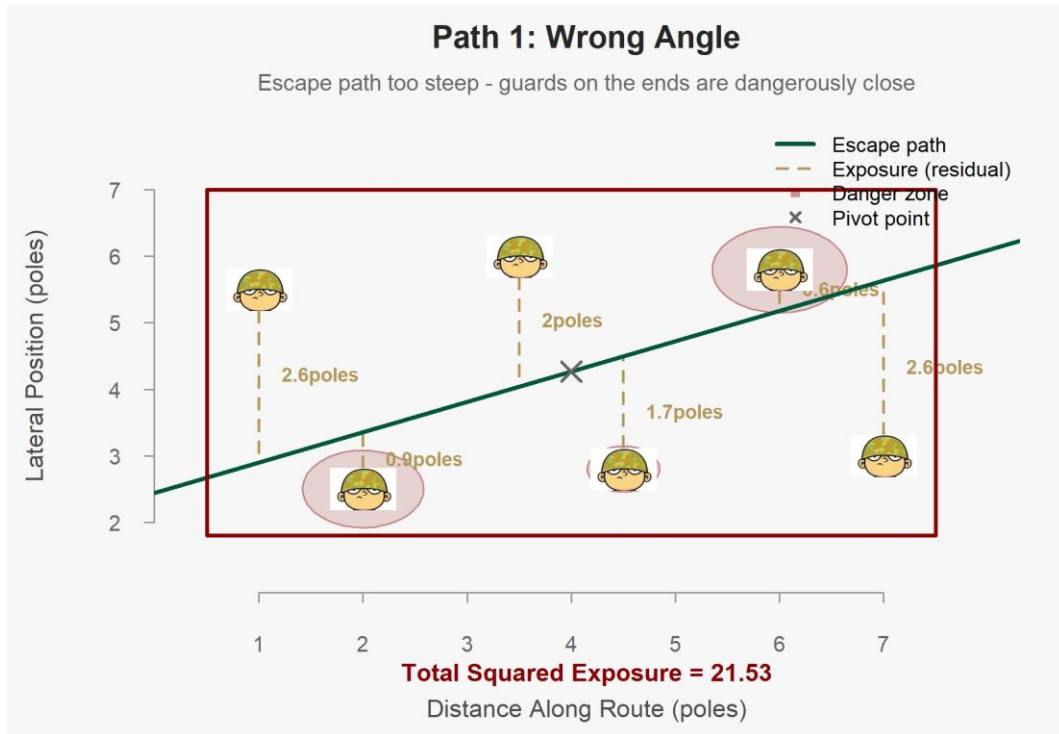


Figure 1

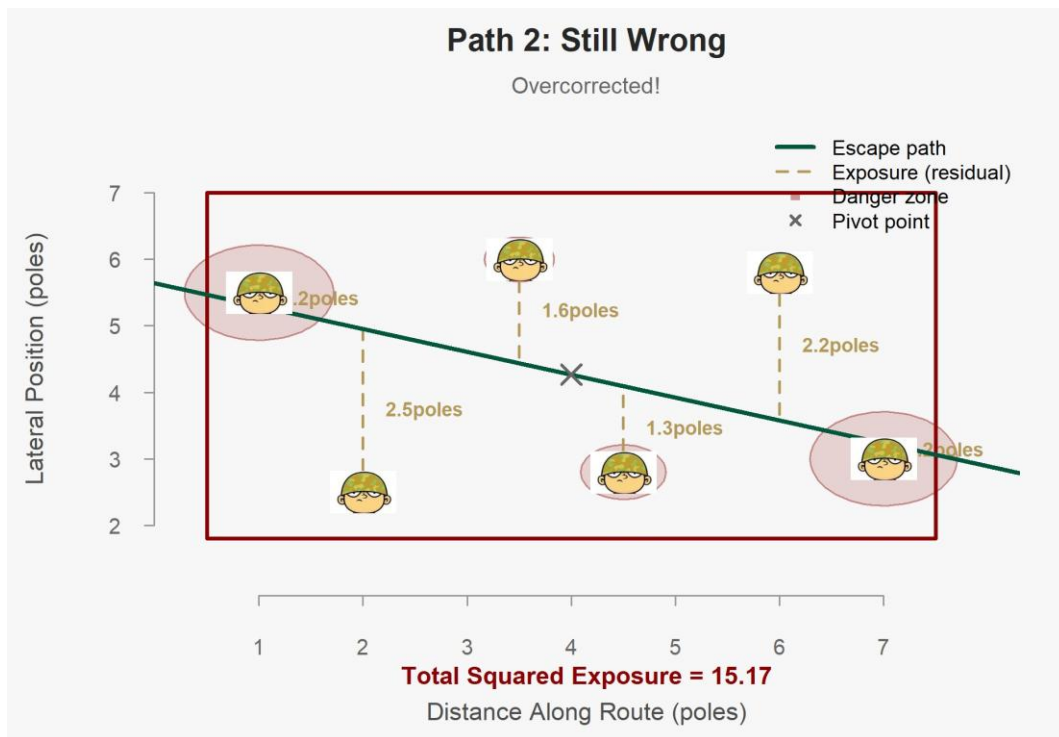


Figure 2

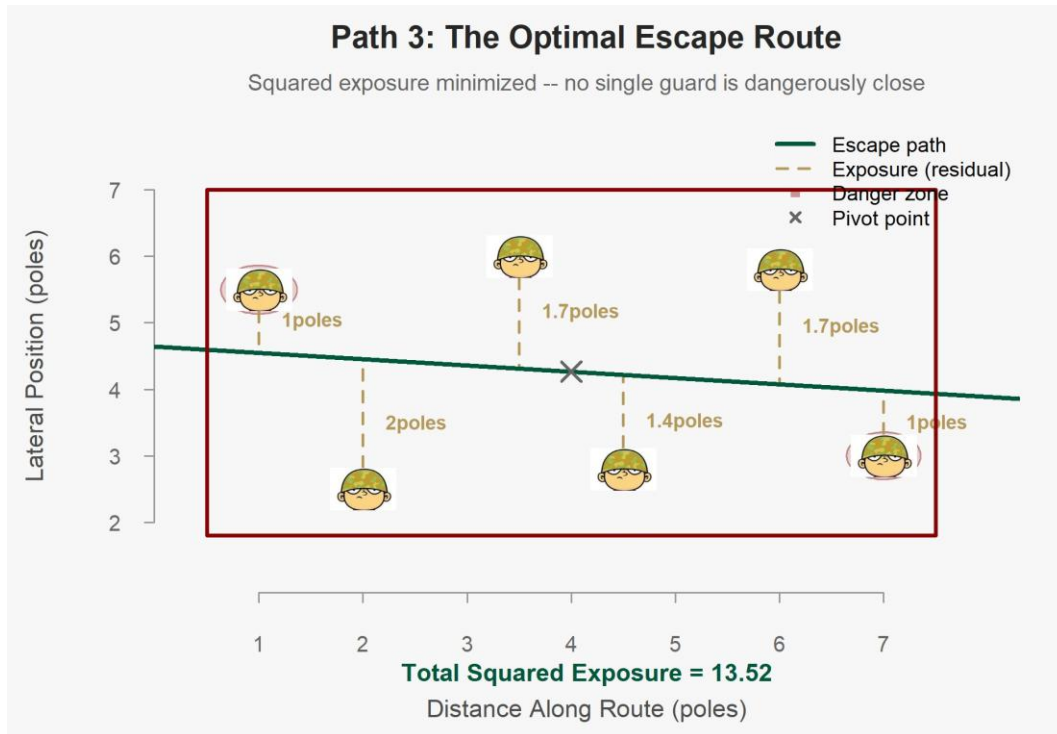


Figure 3

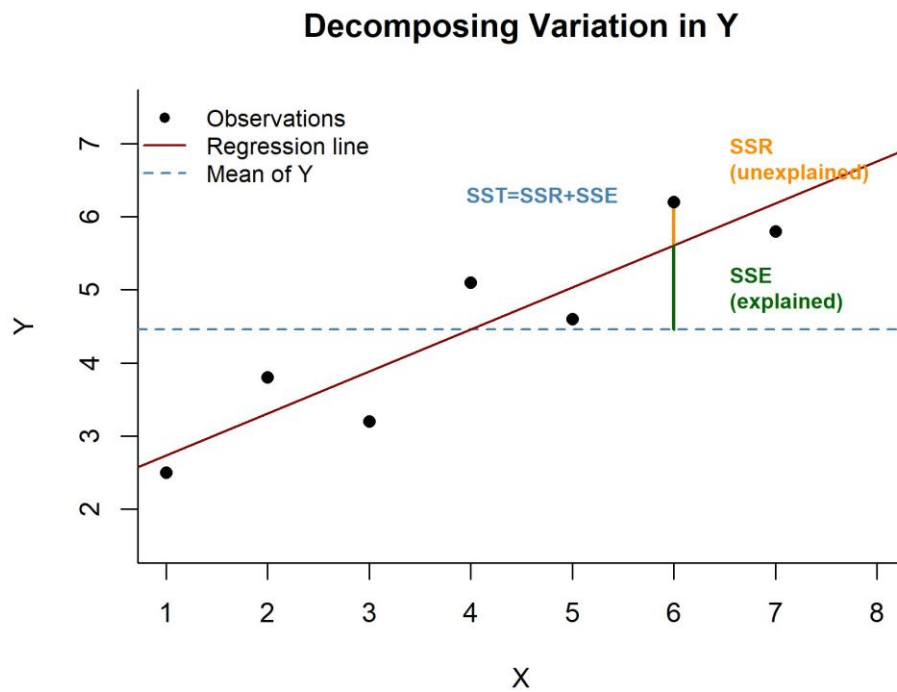


Figure 4: SSR as sum of squared residuals and SSE and sum of squares explained. Plot is representative of concepts utilizing different data.